SECSR2043 OPERATING SYSTEMS
[20 Marks]

Name	:	CHERYL CHEONG KAH VOON	Marks
Student ID	:	A23CS0060	
Section	:	02	

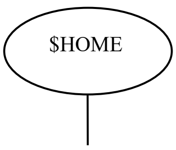
Instruction: Please answer all of the following questions. Whenever the ✋ symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname*.sh (Example: `ahmad.sh`).

```
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*.* .; cp ../fruits/*.* .;
cp ../*.jar .
```

a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with **`$HOME`** directory.

[4 marks

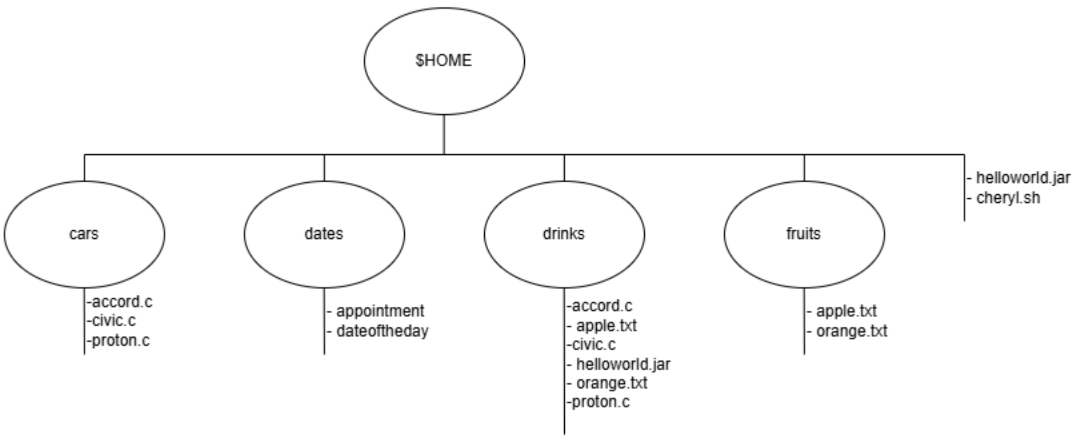


Print screen the script that you type;

```
  GNU nano 7.2                              cheryl.sh
echo  ^`^|Hello world ^`^} > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo  ^`^|Honda Accord ^`^} > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*.* .; cp ../fruits/*.* .;
cp ../*.jar .
```

Then draw the tree

```
cheryl@secr2043:~$ tree
.
|-- cars
|   |-- accord.c
|   |-- civic.c
|   `-- proton.c
|-- cheryl.sh
|-- dates
|   |-- appointment
|   `-- dateoftheday
|-- drinks
|   |-- accord.c
|   |-- apple.txt
|   |-- civic.c
|   |-- helloworld.jar
|   |-- orange.txt
|   `-- proton.c
|-- fruits
|   |-- apple.txt
|   `-- orange.txt
`-- helloworld.jar
```



b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display c program files, and enter "c" as the input to the bash script. [4 marks]

Print screen the bash script you type and run

```
  GNU nano 7.2                              ifiles.sh
#!/bin/bash

#prompt the user to enter a file extension
read -p "Enter the file extension (without the dot): " ext

#Find and display all files with the given extension
echo "Files with .${ext} extension:"
find . -type f -name  "*.${ext}"

#count the number of files with the given extension
count =$(fing .type f -nmae "*.${ext}" |wc -1
echo "Number of files with .${ext} extension: $count"
```

```
cheryl@secr2043:~$ nano ifiles.sh
cheryl@secr2043:~$ chmod u+x ifiles.sh
cheryl@secr2043:~$ ./ifiles.sh
Enter the file extension (without the dot): c
Files with .c extension:
./drinks/civic.c
./drinks/accord.c
./drinks/proton.c
./cars/civic.c
./cars/accord.c
./cars/proton.c
Number of files with .c extension: 6
```

2. The following Figure 1 illustrates a tree structure of some directories and files.

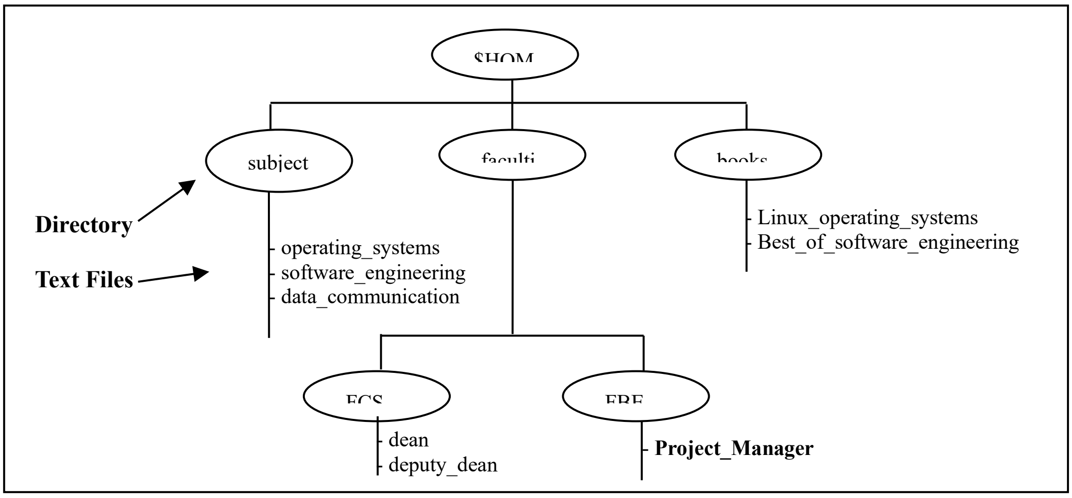


Figure 1

a) Write a bash script (called `myname2a.sh`) that will produce directories and files as in Figure 1. Each text files contain its filename without the underscore character**.** For example: text file `Project_Manager` contains `Project Manager`). [4 marks]

Print screen the bash script you type and run

```
  GNU nano 7.2                                    cheryl2a.sh
#!/bin/bash

#Create directories
mkdir -p $HOME/subjects $HOME/faculties/FCS $HOME/faculties/FBE $HOME/books

#Create and write to text files in the subject directory
echo "operating system" >$HOME/subject/operating_systems
echo "software engineering" >$HOME/subjects/software_engineering
echo "data communication" >$HOME/subjects/data_communication

#Create and write to text files in the faculties/FCS directory
echo "dean" >$HOME/faculties/FCS/dean
echo "deputy dean" >$HOME/faculties/FCS/deputy_dean

#Create and write to text files in the faculties/FBE directory
echo "Project Manager" >$HOME/faculties/FBE/Project_Mananger

#Create and write to text files in the books directory
echo "Linux operating system" >$HOME/books/Linux_operating_systems
echo "Best of software engineering" >$HOME/books/Best_of_software_engineering
```

```
cheryl@secr2043:~$ nano cheryl2a.sh
cheryl@secr2043:~$ chmod u+x cheryl2a.sh
cheryl@secr2043:~$ ./cheryl2a.sh
cheryl@secr2043:~$ tree $HOME
/home/cheryl
|-- books
|   |-- Best_of_software_engineering
|   `-- Linux_operating_systems
|-- cars
|   |-- accord.c
|   |-- civic.c
|   `-- proton.c
|-- cheryl.sh
|-- cheryl2a.sh
|-- dates
|   |-- appointment
|   `-- dateoftheday
|-- drinks
|   |-- accord.c
|   |-- apple.txt
|   |-- civic.c
|   |-- helloworld.jar
|   |-- orange.txt
|   `-- proton.c
|-- faculties
|   |-- FBE
|   |   `-- Project_Mananger
|   `-- FCS
|       |-- dean
|       `-- deputy_dean
|-- fruits
|   |-- apple.txt
|   `-- orange.txt
|-- helloworld.jar
|-- ifiles.sh
`-- subjects
    |-- data_communication
    |-- operating_systems
    `-- software_engineering

10 directories, 25 files
```

b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for directory called `book`. [2 marks]

Directory/File	Access Control
books	drwxrwxr-x
subjects	drwxrwxr-x
Best_of_software_engineering	-rw-rw-r--
FCS	drwxrwxr-x
project_manager	-rw-rw-r--

```
cheryl@secr2043:~$ ls -ld books
ls -ld subjects
ls -ld faculties/FCS
ls -l books/Best_of_software_engineering
ls -l faculties/FBE/Project_Mananger
drwxrwxr-x 2 cheryl cheryl 4096 Jun 16 10:14 books
drwxrwxr-x 2 cheryl cheryl 4096 Jun 16 10:14 subjects
drwxrwxr-x 2 cheryl cheryl 4096 Jun 16 10:14 faculties/FCS
-rw-rw-r-- 1 cheryl cheryl 29 Jun 16 10:14 books/Best_of_software_engineering
-rw-rw-r-- 1 cheryl cheryl 16 Jun 16 10:14 faculties/FBE/Project_Mananger
```

c) Write another bash script (called `myname2c.sh`) that will change the access control of the directories and files based on the following information:

[4 marks]

Directory/File	Users								
	Owner			Group			Public		
subjects	✓	✓	✓	✓	x	x	✓	x	x
Best_of_software_engineering	✓	x	✓	x	✓	x	x	x	x
FCS	✓	✓	x	x	x	x	✓	✓	✓
project_manager	x	x	x	x	✓	✓	x	x	✓

Print screen the bash script you type and run

```
  GNU nano 7.2                                      cheryl2c.sh
#!/bin/bash

#Set permission for subjects directory
chmod 744 $HOME/subjects

#Set permission for Best_of_software_engineering file
chmod 520 $HOME/books/Best_of_software_engineering

#Set permission for FCS directory
chmod 607 $HOME/faculties/FCS

#Set permission for project_mananger file
chmod 031 $HOME/faculties/FBE/Project_Mananger
```

```
cheryl@secr2043:~$ nano cheryl2c.sh
cheryl@secr2043:~$ chmod +x cheryl2c.sh
cheryl@secr2043:~$ ./cheryl2c.sh
cheryl@secr2043:~$ ls -ld ~/subjects
ls -ld ~/faculties/FCS
ls -l ~/books/Best_of_software_engineering
ls -l ~/faculties/FBE/Project_Mananger
drwxr--r-- 2 cheryl cheryl 4096 Jun 16 10:14 /home/cheryl/subjects
drw----rwx 2 cheryl cheryl 4096 Jun 16 10:14 /home/cheryl/faculties/FCS
-r-x-w---- 1 cheryl cheryl 29 Jun 16 10:14 /home/cheryl/books/Best_of_software_engineering
-----wx--x 1 cheryl cheryl 16 Jun 16 10:14 /home/cheryl/faculties/FBE/Project_Mananger
```

d) Complete the following table by writing the access control for each directory or file after executing the bash script in question 2(c)). [2 marks]

Directory/File	Access Control
subjects	drwxr--r--
Best_of_software_engineering	drw----rwx
FCS	-r-x-w----
project_manager	-----wx--x

End of Lab 3

**** All the Best for Final Exam ****